Ch. 15 CW #2

Principles of Microeconomics

December 9, 2025

Instructions

Please answer the following on a separate sheet of paper, fully justify your answers, and turn your answers in on Gradescope by the beginning of next class.

15.2. The Competitive Firm's Supply Curve

1. Under what conditions will a firm shut down temporarily? Explain.
2. The state of Maine has a very active lobster industry, which harvests lobsters during the summer months. During the rest of the year, lobsters can be obtained from other parts of the world but at a much higher price. Maine is also full of "lobster shacks," roadside restaurants serving lobster dishes that are open only during the summer. Explain why it is optimal for lobster shacks to operate only during the summer.
3. Under what conditions will a firm exit a market? Explain.
4. Below are profit-maximizing rules for a competitive firm. Fill in the blank.
   (a) Find Q at which $P =$ ____.
   (b) If $P <$ _____, shut down immediately and remain out of business.
   (c) If _____ $< P <$ _____, operate in the short run but exit in the long run.
   (d) If _____ $< P$, stay in business and enjoy your profits!
5. Draw a short-run diagram showing a U-shaped average total cost curve, a U-shaped average variable cost curve, and a U-shaped marginal cost curve. On it, indicate the range of output and the range of price for which the following actions are optimal.
   (a) The firm shuts down immediately.
   (b) The firm operates in the short run despite sustaining a loss.
   (c) The firm operates while making a profit.

15.3. The Competitive Market's Supply Curve

1. Which of the following events will induce firms to enter an industry? Which will induce firms to exit? When will entry or exit cease? Explain your answer.

   (a) A technological advance lowers the fixed cost of production of every firm in the industry.

   (b) The wages paid to workers in the industry go up for an extended period of time.

   (c) The price of a key input rises due to a long-term shortage of that input.

2. Assume that the egg industry is perfectly competitive and is in long-run equilibrium with a perfectly elastic long-run industry supply curve. Health concerns about cholesterol then lead to a decrease in demand. Draw a series of figures like those in section 15.E. to show the short-run behavior of the market and how long-run equilibrium is reestablished.